

Camilo A. Cárdenas-Hurtado

CONTACT INFORMATION

Room: COL.8.05
Department of Statistics
London School of Economics

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RESEARCH INTERESTS

Statistical machine learning with latent variables, Psychometrics, IRT, diagnostic models, distributional and semi-parametric regression, high-dimensional categorical data analysis, causal inference, applications in educational testing, psychology, political science, economics.

ACADEMIC APPOINTMENTS

London School of Economics and Political Science (LSE), London, UK.

LSE Fellow

2023/09 -

EDUCATION

London School of Economics and Political Science (LSE), London, UK.

PhD, STATISTICS (Social Statistics), 2023

- Thesis: *Generalised Latent Variable Models for Location, Scale, and Shape Parameters* (Supervisors: [Prof. Irini Moustaki](#) and [Dr. Yunxiao Chen](#))

Universidad Nacional de Colombia (UNAL), Bogotá D.C., Colombia.

MSc, STATISTICS, 2017

Graduate Diploma (Specialization), STATISTICS, 2014

BA, ECONOMICS, 2011

PEER-REVIEWED PUBLICATIONS

Cárdenas-Hurtado, C.A., Moustaki, I., Chen, Y., Marra, G. (2025) Generalized Latent Variable Models for Location, Shape, and Scale parameters. *Psychometrika*, 90(3):932-956 ([Link](#), R package: [glvmlss](#)).

Cárdenas-Hurtado, C.A., Hernández, M.A., Torres, J.E. (2015), A Statistical Analysis of Heterogeneity on Labour Markets and Unemployment Rates in Colombia. *Revista Desarrollo y Sociedad*, 75(1):153-196, ([Link](#))

PREPRINTS AND WORK IN PROGRESS

A Generalized Additive Partial-Mastery Cognitive Diagnostic Model, with Y. Chen, I. Moustaki. *Major revisions at The Annals of Applied Statistics*. (arXiv preprint: [Link](#), R package: [gapmCDM](#))

An L_1 Rotation Method for Exploratory Bi-factor Analysis, with T.G Vasiljev, Y. Chen, and I. Moustaki (*In preparation*, R package: [bifacLlrot](#))

Assessing DIF on Asymmetric Item Response Theory Models with Bounded Continuous Responses, with D. Molenaar (*In preparation*)

Ordering consistency: Generating indexes using ordinal data for regression analysis, with C. F. Balcázar (*In preparation*)

POLICY PAPERS

Cárdenas-Hurtado, C.A., Hernández, M.A. (2019), Understanding the Consumer Confidence Index in Colombia: A structural FAVAR analysis. *Policy paper series - Borradores de Economía*, No. 1063. ([Link](#))

Cárdenas-Hurtado, C.A., Garavito-Acosta, A., Toro-Córdoba, J. (2018), Asymmetric Effects of Terms of Trade Shocks on Tradable and Non-tradable Investment Rates: The Colombian Case. *Policy paper series - Borradores de Economía*, No. 1043. ([Link](#))

Velasco, A.M., **Cárdenas-Hurtado, C.A.** (2015), A Macro CGE Model for the Colombian Economy. *Policy paper series - Borradores de Economía*, No. 863. ([Link](#))

ACADEMIC TALKS

2026: COMPS Seminar (invited talk, Institute of Computer Science, Czech Academy of Sciences, Prague, CZ); Department of Statistics' Research Showcase (LSE); International Symposium on Nonparametric Statistics (ISNPS26, Thessaloniki, GR); International Meeting of the Psychometric Society (IMPS26, contributed, Seoul, KR).

2025: Frontier Research in Educational Measurement (FREMO25, contributed, Oslo, NO); 15th Scientific Meeting of the Classification and Data Analysis Group (CLADAG25, invited speaker, Naples, IT).

2024: Brown-bag Statistics Seminar (online invited talk, Pennsylvania State University); International Meeting of the Psychometric Society (IMPS24, contributed, Prague, CZ).

2023: Social Statistics Seminar (LSE); Modern Modeling Methods (M3, contributed, CT, US); International Meeting of the Psychometric Society (IMPS23, contributed, MD, US).

2022: International Meeting of the Psychometric Society (IMPS22, contributed, travel award, Bologna, IT).

2021: Royal Statistical Society International Conference (RSS21, invited symposium, Manchester, UK).

TEACHING EXPERIENCE

London School of Economics and Political Science (LSE)

- DEPARTMENT OF STATISTICS
 - ST102: Elementary Statistical Theory
 - ST201: Statistical Models and Data Analysis
 - ST205: Sample Surveys and Design of Experiments
 - ST211: Applied Regression
 - ST309: Elementary Data Analytics (Lecturer)
 - ST312: Applied Statistics Project
 - ST405/MY455: Multivariate Methods
 - ST416: Multilevel Modelling

SCHOLARSHIPS & AWARDS

Class Teacher Award (LSE): 2022, 2024, 2025.

Psychometric Society Travel Award (Best student paper): 2022.

MPhil/PhD Scholarship (LSE): 2019-2023.

ACADEMIC
SERVICE

London School of Economics and Political Science (LSE)

- Academic Mentor, PhD representative, member of the Equity, Diversity and Inclusion Committee (EDI) and Departmental Teaching Committee (DTC), Mentor for MSc students.

Psychometric Society

- Member of the Diversity, Equity and Inclusion (DEI) Committee (2025-2028).

Editorial work

- Reviewer for: British Journal of Mathematical and Statistical Psychology (x1), Journal of Multivariate Analysis (x1), Psychological Methods (x1), Statistical Papers (x1).

PROFESSIONAL
EXPERIENCE

Banco de la República (Central Bank of Colombia), Bogotá, Colombia.
Macroeconomic Programming and Inflation Department.

Specialist Economist
Economist

2015/03 - 2019/07
2011/08 - 2015/02

SKILLS

Languages: Spanish (native), English (fluent), French (basic)

Programming: Proficient in: R, C++ (Rcpp), Python, Stata, MATLAB
Working Knowledge: Julia, GAMS

REFERENCES

Professor Irini Moustaki e-mail: i.moustaki@lse.ac.uk
Department of Statistics
London School of Economics and Political Science

Dr. Yunxiao Chen e-mail: y.chen186@lse.ac.uk
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